

SOLVING QUADRATIC INEQUALITIES**9.2.1**

There are several methods for solving **quadratic inequalities** in one variable, but one method that works for many kinds of inequalities is to change the inequality to an equation, solve it, and then graph the solution(s) on a number line. The solution(s) to the equation, called **boundary point(s)**, divide the number line into regions. By checking a number from each region in the **original** inequality it can be determined if the numbers in that region are solutions. The boundary points are included in ($\geq$ or $\leq$) or excluded from ($>$ or $<$) the solution depending on the inequality sign.

For additional information see the Math Notes box in Lesson 9.2.2.

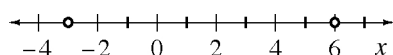
Example 1 Solve: $x^2 - 3x - 18 < 0$

Change to an equation and solve.

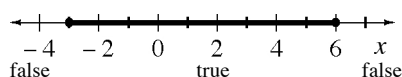
$$x^2 - 3x - 18 = 0$$

$$(x - 6)(x + 3) = 0$$

$x = 6$ or $x = -3$ (the boundary points)



Choosing $x = -4, 0, 7$ to test in the original inequality, $x = -4$ is false, $x = 0$ is true, and $x = 7$ is false.



The solution is all numbers greater than -3 and less than 6 , written as $-3 < x < 6$.

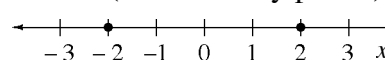
Example 2 Solve: $m^2 - 3 \geq 1$

Change to an equation and solve.

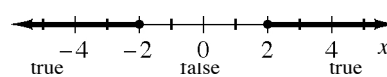
$$m^2 - 3 = 1$$

$$m^2 = 4$$

$m = \pm 2$ (the boundary points)



Choosing $m = -3, 0, 3$ to test in the original inequality, $m = -3$ is true, $m = 0$ is false, and $m = 3$ is true.



The solution is all numbers less than or equal to -2 or all numbers greater than or equal to 2 , written as $m \leq -2$ or $m \geq 2$.

Problems

Solve each inequality.

1. $x^2 + 6x + 8 < 0$

2. $y^2 - 5y > 0$

3. $y^2 - 5y < 0$

4. $x^2 - 3x - 4 < 0$

5. $-x^2 - 9x - 14 < 0$

6. $y^2 \leq 16$

7. $3x^2 + 7x - 6 \geq 0$

8. $x^2 + 4x - 8 < 4$

9. $y^2 + 6y + 9 > 0$

10. $x(7x - 26) \leq 8$

Answers

- | | | |
|--|-------------------------|----------------------------|
| 1. $-4 < x < -2$ | 2. $y < 0$ or $y > 5$ | 3. $0 < y < 5$ |
| 4. $-1 < x < 4$ | 5. $x < -7$ or $x > -2$ | 6. $-4 \leq y \leq 4$ |
| 7. $x \leq -3$ or $x \geq \frac{2}{3}$ | 8. $-6 < x < 2$ | 9. all numbers except -3 |
| 10. $-\frac{2}{7} \leq x \leq 4$ | | |